

# LECTURE PLAN – COMPUTER ARCHITECTURE

## Lecture 12 of 45

**Topic: Instruction Cycle**

**Unit: Unit 3**

### Learning Outcomes (LO)

- LO1: Define and explain the basic concepts of Instruction Cycle. (L1 – Remember)
- LO2: Explain the working and important features of Instruction Cycle. (L2 – Understand)
- LO3: Apply the concept of Instruction Cycle to simple examples or problems. (L3 – Apply)

### Teaching Content

- Fetch, decode and execute phases
- Register transfers during instruction cycle
- Simple instruction-cycle example

### Teaching–Learning Activities

| Activity      | Description                                                                        |
|---------------|------------------------------------------------------------------------------------|
| Demo          | Teacher explains Instruction Cycle using simple examples on the board/PPT.         |
| Peer Activity | Students discuss a short example in pairs and compare their answers.               |
| Board Problem | Students solve one basic problem based on Instruction Cycle with teacher guidance. |

### Assessment Methods

- Quiz: Check understanding of key terms and concepts.
- Assignment: Solve/write a short problem related to the lecture topic.
- Viva: Explain the main concept and one example in simple words.

### Teaching Aids

- PPT with examples, whiteboard, textbook diagrams, and simple classroom examples.